Javad Komijani

Computational Scientist & ML Researcher

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Professional Summary

Computational scientist and machine learning researcher specializing in generative models, diffusion processes, and normalizing flows on structured and symmetry-constrained domains. Experienced in building PyTorch-based scientific ML frameworks, distributed training systems, and GPU-accelerated simulation pipelines. Focus on scalable generative modeling, geometric deep learning, and scientific computing at scale.

Selected Research Impact

- 26 peer-reviewed publications, 3000+ citations, h-index 19.
- Contributions to precision determinations of parameters of the Standard Model of Particle Physics.
- Developed frameworks for controlling multi-scale expansions in quantum field theory.
- Introduced methods for handling divergent perturbative series in physics computations.
- Developed an implicit score-matching framework for Lie groups.
- Developed GPU-accelerated software frameworks for diffusion models and normalizing flows.

Research & Technical Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*
- Developed diffusion and score-based generative models on Lie groups for lattice gauge theory.
- Built differentiable computing frameworks using PyTorch (ODE solvers and linear algebra).
- Integrated normalizing flow modules into the EU interTwin Digital Twin Engine.
- Developed distributed training infrastructure for large-scale lattice simulations.
- Supervised PhD, MSc, BSc, and internship projects in scientific machine learning.

2015–2021 **Postdoctoral Researcher**, *TU Munich / Glasgow / Tehran*
- Monte Carlo simulations for lattice QCD observable.
- Bayesian parameter estimation from Lattice QCD simulations.
- Developed scalable lattice-QCD simulation pipelines in HPC environments.
- Collaborated internationally in computational and mathematical physics projects.

Open-Source Scientific ML Software

lattice_ml **Installable Python package for scientific machine learning**
- Diffusion-based generative modeling on Lie groups and gauge theories.
- Adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.
- Differentiable SVD and eigendecomposition of unitary matrices.

normflow Installable Python package for normalizing flows on Lie groups and lattice gauge theory, including symmetry-preserving transformations and flow-based sampling methods.

interTwin Contributed normalizing-flow-based ML modules to the EU interTwin Digital Twin Engine.

Mentoring & Leadership

Mentoring **Co-supervised PhD, MSc, and BSc researchers in physics & scientific ML:**

- Gauge-equivariant diffusion models for lattice field theory (PhD project).
- Fourier acceleration methods for SU(N) lattice gauge theory (PhD project).
- Gauge-equivariant U-Net for diffusion models in lattice gauge theory (MSc project).
- Autoregressive generative models for lattice configurations (MSc project).
- Continuous-time generative models and normalizing flows (MSc projects).
- Rational-quadratic splines for normalizing flows (MSc projects).
- CNN architectures in arbitrary dimensions for scientific data (Intern project).

Organization **Co-organized workshops on physics & machine learning:**

- ZPW 2026 — Machine Learning Techniques for Particle Physics.
- QCD 2025 — Symposium on QCD Factorization.
- ML Lattice 2025 — Machine learning approaches in Lattice QCD.

Technical Skills

Programming	Python, C++, Cython, Bash
ML/AI	PyTorch, Diffusion Models, Normalizing Flows, Generative Modeling, Distributed Training
Mathematics	Differential Equations, Linear Algebra, Optimization, Statistical Modeling
Tools	Linux, Git, SQL, Pandas
Scientific	NumPy, SciPy, Scikit-learn, HPC, Parallel Computing, Monte Carlo Methods

Education

2015 **PhD in Physics**, *Washington University in St. Louis*

Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*

- Developed EFT-based frameworks and Bayesian inference methods for lattice QCD.
- Performed precision determinations of hadronic observables in lattice QCD.
- Developed a novel definition of eigenvalue problems for nonlinear systems arising in physics.

2009 **MSc in Electrical Engineering**, *University of Tehran*

- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*

Publications

A complete list of over 26 peer-reviewed publications can be found at this link.

Languages

English	Fluent
German	Basic (A2)